



Project Report On

EchoLog: An AI Journal

*Submitted in partial fulfillment of the requirements for the
award of the degree of*

Bachelor of Technology

in

Computer Science and Engineering

By

Alvina Sheen T (U2203033)

Adarsh S (U2203016)

Basil Raju (U2203064)

Under the guidance of

Ms. Shini Sam

**Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)**

Rajagiri Valley, Kakkanad, Kochi, 682039

March 2026

CERTIFICATE

*This is to certify that the project report entitled "**EchoLog: An AI-Journal**" is a bonafide record of the work done by **Alvina Sheen T (U2203033)**, **Adarsh S (U2203016)**, **Basil Raju (U2203064)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

Project Guide

Ms. Shini Sam

Assistant Professor

Dept. of CSE

RSET

Project Co-ordinator

Dr. Jisha G

Associate Professor

Dept. of CSE

RSET

Head of Department

Dr. Preetha K G

Professor

Dept. of CSE

RSET

ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Fr. Dr. Jaison Paul Mulerikkal** CMI, Principal, RSET, and **Dr. Preetha K G**, Head of Department, Department of Computer Science and Engineering, for providing us with the opportunity to undertake our project, “EchoLog:An AI - Journal”.

We are highly indebted to our project coordinators, **Dr. Jisha G**, Associate Professor, Department of Computer Science and Engineering and **Dr. Preetha K. G.** , Head Of Department, Department of Computer Science and Engineering, for their valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express our sincere gratitude to our project guide **Ms. Shini Sam**, Assistant Professor, Department of Computer Science and Engineering, for her patience and all the priceless advice and wisdom she has shared with us.

Last but not the least, we would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Alvina Sheen T

Adarsh S

Basil Raju

Abstract

Digital journaling is commonly used by many people to record daily experiences and to reflect their thoughts and emotions. But most of the traditional journaling applications mainly store written entries and do not give much support for analysing emotional changes over time. Because of this reason, users sometimes find it difficult to understand their emotional patterns and also may not continue journaling regularly.

This project explains the development of **EchoLog**, an AI supported digital journaling platform which allows users to input their thoughts using text as well as voice entry. The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) methods to analyse journal entries and give simple insights about the emotional tone written by the user.

Voice entries are processed using Open Whisper AI, speech recognition model, which convert spoken words into text format. After conversion, the text is analysed using NLP tools such as TextBlob and transformer based DistilBERT model to perform sentiment analysis and emotion detection. With the help of these methods, the application can observe emotional patterns in journal entries and give small reflective feedback based on what the user writes.

EchoLog is designed with a simple interface so that journaling becomes easy to use and maintain. By combining voice journaling, sentimental analysis, keyword based journaling, and emotion detection, the platform helps users to understand their feelings better and also encourages regular self reflection.

Keywords: Artificial Intelligence, Whisper AI, Natural Language Processing, Voice-to-Text Journaling, Mood Detection, Gamified Emotional Intervention

Contents

Acknowledgment	i
Abstract	ii
List of Abbreviations	vi
List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	2
1.3 Scope and Motivation	2
1.4 Objectives	2
1.5 Challenges	3
1.6 Assumptions	3
1.7 Societal / Industrial Relevance	3
1.8 Organization of the Report	4
1.9 Summary of Chapter	4
2 Literature Survey	5
2.1 Paper 1: Li, H., Shi, K., Wang, D., Wang, X., & Xu, G. (2024). Topic- Controllable Keywords-to-Text Generator with Knowledge Base Network .	5
2.2 Paper 2: Liu, Y., Yang, X., & Qu, D. (2024). Exploration of Whisper Fine-Tuning Strategies for Low-Resource ASR	6
2.3 Paper 3: Rijal, S., Adhikari, S., Dahal, M., Awale, M., & Ojha, V. (2024). Whisper Fine-Tuning for Nepali Speech Recognition	6
2.4 Paper 4: MoodSync – Emotion-Aware Mood Detection System	6

2.5	Comparison of Existing Approaches	7
2.6	Research Gaps Identified	8
2.7	Summary of Chapter	8
3	System Design	9
3.1	Overall Architecture	9
3.2	Component Design	10
3.3	Module Division	11
3.4	Tools and Technologies	11
3.5	Work Breakdown and Responsibilities	12
3.6	Key Deliverables	12
3.7	Project Timeline	12
3.8	Summary of Chapter	13
4	Results and Discussions	14
4.1	User Authentication and Account Creation	14
4.2	Mode Selection and Dashboard Interface	15
4.3	Journal Entry Methods	16
4.4	Keyword-Based Journaling	17
4.5	Text-Based Journaling	18
4.6	Voice-to-Text Journaling	19
4.7	Task Management Features	20
4.8	Reminder and Notification System	20
4.9	Stress Relief and Mini-Games	21
4.10	System Performance Evaluation	22
4.10.1	Voice-to-Text Accuracy	22
4.10.2	Sentiment Analysis Performance	23
4.10.3	User Experience Metrics	24
4.10.4	System Performance Metrics	25
5	Conclusions & Future Scope	26
	References	27

List of Publications	27
Appendix A: Presentation	29
Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes	49
Appendix C: CO-PO-PSO Mapping	54

List of Abbreviations

AI - Artificial Intelligence

NLP - Natural Language Processing

STT - Speech-to-Text

ML - Machine Learning

API - Application Programming Interface

UI - User Interface

UX - User Experience

BERT - Bidirectional Encoder Representations from Transformers

List of Figures

3.1	Overall Architecture of EchoLog System	10
3.2	Project Timeline (Gantt Chart)	13
4.1	Login page of the EchoLog system	15
4.2	User registration interface for creating new EchoLog account	15
4.3	Mode selection interface for choosing journaling type	16
4.4	Main dashboard of EchoLog journaling system	16
4.5	Interface showing available journaling methods	17
4.6	Keyword input interface for journaling	17
4.7	Keyword suggestion feature to help user during journaling	18
4.8	Generated journal entry saved after keyword processing	18
4.9	Text journaling interface for entering journal content	18
4.10	Recently stored journal entries display	19
4.11	Voice to text journaling interface using speech recognition	19
4.12	Interface for adding tasks to to-do list	20
4.13	Task added and displayed in task list	20
4.14	Reminder configuration interface	20
4.15	Reminder notification shown to user	21
4.16	Mini games interface for engagement	21
4.17	Memory game for focus and relaxation	22
4.18	Voice to text accuracy evaluation	22
4.19	Precision, recall and F1 score for sentiment classification	23
4.20	User experience evaluation metrics	24
4.21	System performance metrics showing response time and efficiency	25

List of Tables

2.1	Comparison of Key Research Works	7
5.1	CO-PO-PSO Mapping	55

Chapter 1

Introduction

This chapter introduce the concept of AI powered digital journaling and its role in improving personal reflection and emotional awareness. It gives overview of the project background, problem definition, scope and motivation, objectives, challenges, assumptions, relevance in society and organization of report. The main objective of this chapter is to explain the need for a smart journaling platform which combine Artificial Intelligence and Natural Language Processing to analyse user entries and give meaningful emotional insights.

1.1 Background

Journaling has been known as an effective method for self recognition, emotional expression and personal development. Also traditional journaling methods involve writing daily experiences and thoughts in notebooks or simple digital documents. Traditional journaling is mostly passive, that means it only stores the user entry and does not interact with the user. Also there is no proper journaling system which combine voice to text, keyword based journaling, emotion detection, sentimental analysis, to do list and other features in single unified system.

With fast development of Artificial Intelligence (AI) and Natural Language Processing (NLP), it is now possible to create intelligent systems which can analyse text data and interact with user. This AI based journaling platform can analyse journal entries, detect emotions and trigger mini stress relief games if the user tone is sad or negative. Also speech recognition technologies allow users to record journal entries using voice, which makes journaling easier and more accessible.

The proposed system EchoLog combine speech to text technology using Open Whisper AI and apply sentiment analysis and emotion detection using NLP tools like TextBlob and

DistilBERT. By combining these technologies, the system allow users to record thoughts using text or voice and also receive automatic emotional insights from their entries.

1.2 Problem Definition

Traditional journaling platforms mainly focus only on recording text without giving deeper insights about emotions or themes written by users. Because of this, users may find difficult to track their emotional patterns or reflect properly on their experiences. Also many users prefer voice based interaction instead of typing long entries, but normal journaling tools usually do not support voice input.

The main aim of this project is to develop AI powered digital journaling system which allow users to record entries using both text and voice while automatically analysing emotional tone and topics in the entries. The system try to help users in understanding their emotional trends and improve self awareness using intelligent analysis.

1.3 Scope and Motivation

Scope: The scope of this project include development of digital journaling platform which allow users to create journal entries using text and voice input. The system perform speech to text conversion using Whisper and apply Natural Language Processing techniques like TextBlob and DistilBERT to analyse sentiment and detect emotions in journal entries. The platform also store journal entries securely and give users insights about their emotional patterns over time.

Motivation: The motivation of this project is to improve traditional journaling by integrating Artificial Intelligence and Natural Language Processing (NLP). Many people find difficult to keep journaling habit because writing regularly take effort and normal journaling does not give feedback. By adding voice journaling and automatic emotion analysis, the system try to make journaling more engaging, convenient and more user interactive.

1.4 Objectives

- Develop an AI powered digital journaling platform which allow users to record their thoughts and experiences.

- Implement speech to text feature to support voice based journaling.
- Apply Natural Language Processing techniques to analyse sentiment and detect emotions in journal entries.
- Provide useful insights and reflections based on analysed journal content.
- Ensure secure storage and management of journal data.
- Create user friendly interface which encourage regular journaling habit.

1.5 Challenges

One of the main challenge in this project is getting accurate speech to text conversion for voice journal entries. Another challenge is doing correct sentiment and emotion analysis on natural language text, because human emotions are complex and sometimes depend on context. Also keeping user data private and storing journal entries securely is important during system development.

1.6 Assumptions

The project is based on following assumptions:

- Users have device with microphone for recording voice journal entries.
- Internet connection is available for AI model processing and system working.
- Users will enter meaningful journal entries which can be analysed by system.

1.7 Societal / Industrial Relevance

This project has both social and technological relevance. From society side, it help in improving mental well being by encouraging people to reflect on their thoughts and emotions regularly. The system can help teachers, working business professionals and doctors who want to maintain digital journal for emotional expression and memory tracking. From technology side, this project shows practical use of Artificial Intelligence, speech recognition and Natural Language Processing in productivity and mental wellness applications.

1.8 Organization of the Report

The report is organised as follows:

- **Chapter 1: Introduction** – This chapter explain background, problem definition, scope and motivation, objectives, challenges, assumptions, societal relevance and report structure.
- **Chapter 2: Literature Survey** – This chapter discuss existing works related to AI journaling, speech recognition and sentiment analysis.
- **Chapter 3: System Design** – This chapter describe architecture, modules, diagrams, algorithms, software requirements and interface design.
- **Chapter 4: Conclusion and Future Scope** – This chapter give summary of project and possible future improvements.

1.9 Summary of Chapter

This chapter introduced the idea of AI powered digital journaling and explained the need of intelligent tools for emotional reflection and analysis. It discussed the problem, objectives and scope of the project and also showed importance of using AI technologies in journaling platforms. Next chapters will explain literature survey, system design, implementation and results of the proposed EchoLog system.

Chapter 2

Literature Survey

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Automatic Speech Recognition (ASR) have improved the capabilities of intelligent applications in recent years. These technologies enable systems to process human language, generate meaningful text, and detect emotions from user inputs, but still some limitations are present in real usage. Modern journaling systems can use these technologies to provide more smarter and interactive experiences when compared to traditional diary systems. This chapter reviews the key research works which forms the base for the proposed EchoLog system.

2.1 Paper 1: Li, H., Shi, K., Wang, D., Wang, X., & Xu, G. (2024). Topic-Controllable Keywords-to-Text Generator with Knowledge Base Network

Li et al. (2024) proposed a Topic-Controllable Keywords-to-Text (TC-K2T) generation model that convert unordered keywords into meaningful sentences. The system uses a Knowledge Base Network (KBN) along with Knowledge-guided Conditional Variational Autoencoder (KCVAE) to generate context-aware text. The model ensures that the generated sentences follows a specific topic and also maintain relation with the provided keywords.

The main contribution of this research is the ability to generate natural language sentences from structured keyword inputs while keeping topic consistency. This approach is useful in applications such as journaling systems where users may give keywords instead of writing full sentences. However, the system requires large datasets and high computational resources for training the deep learning model, which sometimes makes the implementation difficult.

2.2 Paper 2: Liu, Y., Yang, X., & Qu, D. (2024). Exploration of Whisper Fine-Tuning Strategies for Low-Resource ASR

Liu et al. (2024) investigated different fine-tuning strategies for OpenAI Whisper Automatic Speech Recognition (ASR) model to improve transcription accuracy in low-resource languages. The study evaluated different techniques including vanilla fine-tuning, parameter-specific fine-tuning, and lightweight module-based fine-tuning.

Experimental results showed that fine-tuning reduces the Word Error Rate (WER) by large amount, giving an average improvement of about 56.94%. The research highlights that parameter-efficient fine-tuning methods can improve transcription performance while also reducing the computational cost. Even after these improvements, the model still depends on high-quality training dataset and may face problems when audio contains noise.

2.3 Paper 3: Rijal, S., Adhikari, S., Dahal, M., Awale, M., & Ojha, V. (2024). Whisper Fine-Tuning for Nepali Speech Recognition

Rijal et al. (2024) focused on improving speech-to-text transcription accuracy for Nepali language using the Whisper model. The research combined publicly available Automatic Speech Recognition datasets with newly recorded audio samples and also applied data augmentation techniques to increase dataset diversity.

The study showed that fine-tuning Whisper improved transcription accuracy, reducing the Word Error Rate by 36.2% for the small model and 23.8% for the medium model. The research shows the importance of dataset quality and diversity while training speech recognition systems. However, the model still faces challenges related to accent variation, background noise, and informal speech patterns which can affect the performance.

2.4 Paper 4: MoodSync – Emotion-Aware Mood Detection System

The MoodSync system focuses on detecting and analysing user emotions using Natural Language Processing techniques. The system analyzes textual input from users and identify emotional states such as happiness, sadness, anger, or neutral. Sentiment analysis techniques and machine learning models are used to classify the emotional tone of the

input text.

The main contribution of this work is the integration of emotion detection into user-interactive systems. By identifying emotional states, the system can respond accordingly and improve user engagement and well-being. However, emotion detection models sometimes struggle with contextual meaning, sarcasm, and different writing styles, which may affect the classification accuracy.

2.5 Comparison of Existing Approaches

Table 2.1 compares the major research works discussed in this chapter. The reviewed literature presents different techniques related to text generation, speech recognition, and emotion detection. Each approach focuses on one specific component required for building intelligent interactive systems. This comparison helps in finding the strengths, limitations, and research gaps which motivates the development of the proposed EchoLog system.

Table 2.1: Comparison of Key Research Works

Research Work	Main Technique	Application Area	Key Contribution	Limitations
Li et al. (2024)	Keyword-to-text generation using KBN and KCVAE	Automated text generation	Generates coherent sentences from unordered keywords	Requires large datasets and high computational resources
Liu et al. (2024)	Whisper model fine-tuning for ASR	Speech-to-text transcription	Reduces Word Error Rate for low-resource languages	Depends on high-quality training datasets
Rijal et al. (2024)	Dataset augmentation and Whisper fine-tuning	Speech recognition systems	Improves transcription accuracy for Nepali speech	Sensitive to accents and background noise
MoodSync System	Sentiment and emotion classification using NLP	Emotion detection systems	Identifies emotional states from textual input	Difficulty in handling sarcasm and contextual meanings

2.6 Research Gaps Identified

From the reviewed literature, some research gaps can be identified:

1. Most existing research focus on individual components such as text generation, speech recognition, or emotion detection instead of integrating them into single system.
2. Keyword-to-text generation models generate meaningful sentences but do not consider emotional context or mood detection.
3. Speech recognition systems provide accurate transcription but still struggle with noisy environments and accent variations.
4. Emotion detection systems analyze user mood but usually do not integrate voice input or automated journaling features.
5. There is lack of unified systems which combine keyword-based journaling, voice-to-text transcription, emotion detection, and user engagement mechanisms together.

2.7 Summary of Chapter

This chapter reviewed the major research works related to keyword-based text generation, speech recognition, and emotion detection systems. The comparison shows the strengths and limitations of existing approaches and also helps to identify research gaps in integrating these technologies. These observations gives the base for the proposed EchoLog system, which combines keyword-based sentence generation, voice-to-text journaling using Whisper, and emotion-aware interaction in one intelligent platform.

Chapter 3

System Design

The system design phase explains how the proposed EchoLog AI powered journaling system is structured and how different components work together. The main goal of this design is to combine speech recognition, natural language processing and interactive features to create a smart journaling platform. Users can record their thoughts by typing or by speaking and the system analyses the emotional tone of journal entries. Based on detected sentiment, the application gives supportive response to the user.

The architecture includes different modules like voice to text conversion, sentiment and emotion analysis, journal storage and user engagement features such as reminders, to do list and small mini games. All these parts are connected together to give simple and user friendly journaling experience which helps user to reflect their feelings and also maintain daily productivity.

3.1 Overall Architecture

Figure 3.1 shows overall architecture of EchoLog. The system is designed in modular way so that each function can work separately but still connected with main application. The architecture is divided into four layers which are user interaction layer, speech processing layer, natural language processing layer and application service layer.

In user interaction layer, the user can enter journal entries through voice recording or by typing text. Speech processing layer converts voice input into text using Whisper speech recognition model.

After conversion, the text is passed into natural language processing layer where sentiment analysis and emotion detection is done using TextBlob and DistilBERT. These models help the system to understand emotional meaning of journal entry.

The final layer manages the database and other system services. This includes storing

journal entries, reminder management, to do list and mini games which helps to increase user engagement.

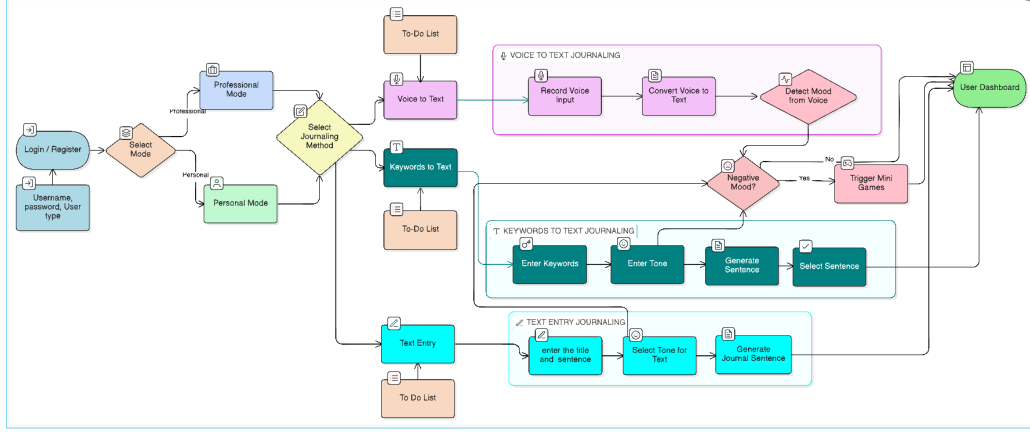


Figure 3.1: Overall Architecture of EchoLog System

3.2 Component Design

EchoLog system is made using several connected components which work together to provide intelligent journaling features. These components includes voice processing module, natural language processing module, journal management system and user engagement features.

Voice processing component converts audio recordings into text using Whisper speech recognition model. This makes easier for user to record thoughts without typing.

Natural language processing component analyse the generated text to find emotional tone. TextBlob is used for basic sentiment classification while DistilBERT is used for more accurate emotion detection.

Journal management component stores all journal entries in database. Entries are saved with date so user can review past thoughts anytime.

User engagement component provide extra features like reminders, to do list and mini games. These features are added to make application more interactive and to encourage user to use journal regularly.

3.3 Module Division

To make system easy to develop and manage, the project is divided into three main modules.

Module 1: Voice to Text Processing Module

This module is responsible for converting voice input into text. Whisper speech recognition model is used to process audio recording and generate text output. Audio is captured through application interface, converted into text and then send to NLP module for further processing. This module makes system more easy for users who prefer speaking instead of typing.

Module 2: Natural Language Processing and Emotion Analysis Module

This module analyse text from journal entries and identify sentiment and emotion. TextBlob is used to classify text as positive, negative or neutral.

To improve accuracy, DistilBERT model is also used because it understand context better than simple sentiment tools. By combining both methods, system gives more reliable emotion detection.

Based on result, the system can give supportive message or suggestion to the user.

Module 3: Journal Management and User Engagement Module

This module manage storage of journal entries and also provide extra features to keep user active in application.

All journal entries are stored in database and arranged by date. This allow users to read previous entries and observe emotional changes over time.

Extra features like reminder notification, to do list management and mini games are included to improve user experience and reduce stress.

3.4 Tools and Technologies

EchoLog system is developed using different software tools and machine learning technologies.

- **Software Technologies**

Python is used as main programming language for backend development. Whisper model is used for speech to text conversion. TextBlob and DistilBERT are used for

sentiment analysis and emotion detection.

User interface is developed using HTML, CSS and JavaScript to create interactive journaling platform. Journal entries and user data are stored in database for proper data management.

- **Development Tools**

Visual Studio Code is used for coding and debugging. GitHub is used for version control and sharing code between team members.

3.5 Work Breakdown and Responsibilities

The project was done as team work and each member worked on different part of system.

- **Member 1:** Developed Natural Language Processing module using TextBlob and DistilBERT for sentiment and emotion detection.
- **Member 2:** Implemented a Voice-to-Text module using the OpenAI Whisper speech recognition model to transcribe user voice inputs into text, and performed sentiment and mood detection using DistilBERT and TextBlob.
- **Member 3:** Integrated all modules and developed extra features like reminders, to do list and mini games.

3.6 Key Deliverables

Main deliverables of this project includes EchoLog AI journaling system, voice to text transcription using Whisper, emotion analysis using NLP techniques and user engagement features like reminders, to do list and mini games.

Other deliverables include documentation, architecture diagrams and result analysis showing working of system.

3.7 Project Timeline

Figure 3.2 shows project timeline. The development followed steps like problem identification, literature survey, system design, implementation, testing and documentation.

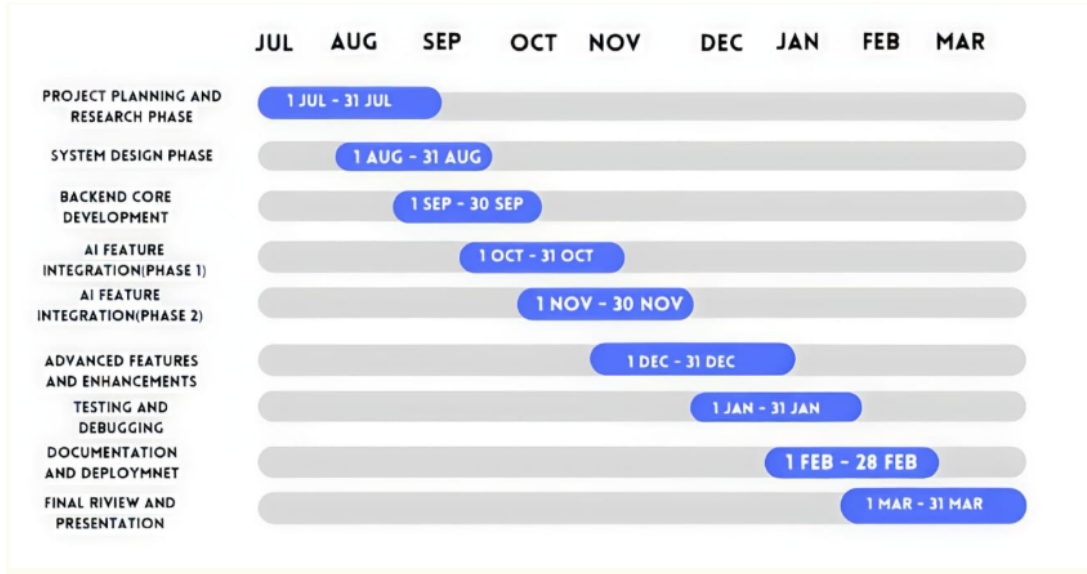


Figure 3.2: Project Timeline (Gantt Chart)

3.8 Summary of Chapter

In this chapter, system design of EchoLog AI journaling platform is explained. Architecture, modules, components and tools used in project are discussed. Roles of team members and project timeline also described.

This design gives base for implementation which is explained in next chapter.

Chapter 4

Results and Discussions

This chapter present the results obtained after implementing the proposed system EchoLog, an AI powered journaling application designed to help users in recording thoughts, analysing emotions and also improving daily productivity. The system was tested using different input methods including text journaling, keyword based sentence generation and voice to text journaling. The evaluation mainly focused on checking the working of mood detection, journaling features, task management, reminder notifications and user engagement tools.

Different user interface components and system modules was tested to make sure smooth interaction between frontend and backend part. The system was also checked for its ability to process user inputs correctly and generate meaningful response. Sentiment analysis was performed on journal entries to find the emotional state of user and give supportive feedback when needed.

The experimental results shows that the system successfully perform sentiment analysis, sentence generation and productivity assistance as expected. The result also shows that combining artificial intelligence techniques with journaling features can improve user engagement and give more interactive journaling experience.

4.1 User Authentication and Account Creation

The system provide secure authentication interface for users to create account and login into EchoLog platform. The registration page allow new users to enter required details and create personal account for using journaling features. After successful registration, users can login into system using their credentials and access main dashboard.

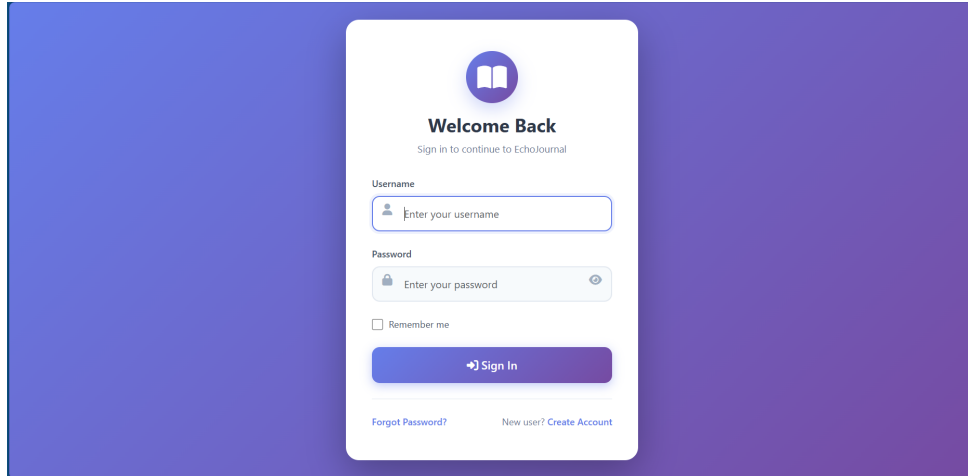


Figure 4.1: Login page of the EchoLog system

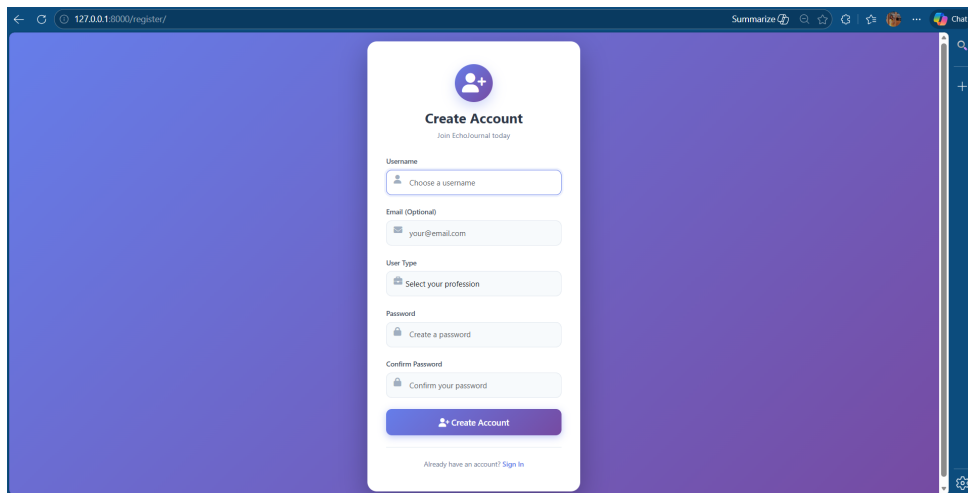


Figure 4.2: User registration interface for creating new EchoLog account

4.2 Mode Selection and Dashboard Interface

After login, user is directed to mode selection interface where they can choose between personal journaling mode and professional journaling mode. This feature helps system to organize journal entries based on user context.

The main dashboard give access to different journaling methods and also extra features like task management, reminders and mini games which are added to improve user engagement.

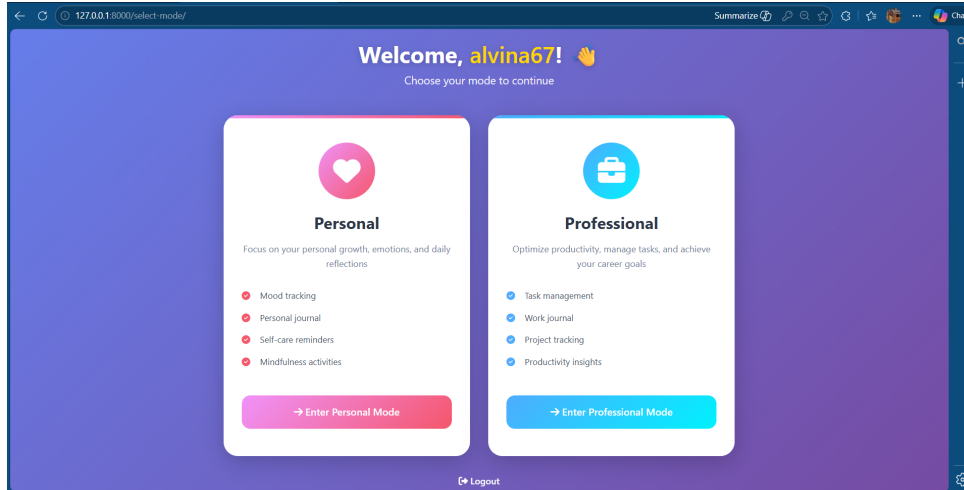


Figure 4.3: Mode selection interface for choosing journaling type

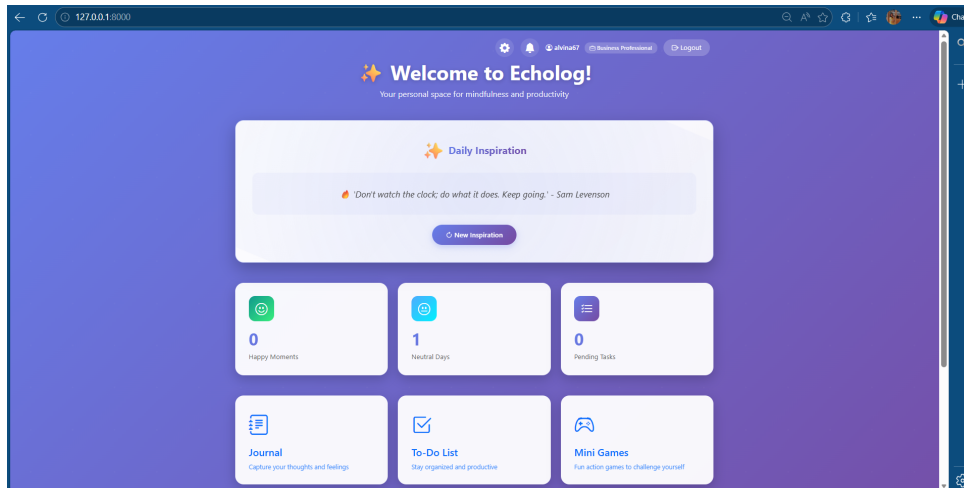


Figure 4.4: Main dashboard of EchoLog journaling system

4.3 Journal Entry Methods

The EchoLog system provide multiple journaling methods including keyword journaling, text journaling and voice to text journaling. This flexibility allow users to choose convenient way to record their thoughts.

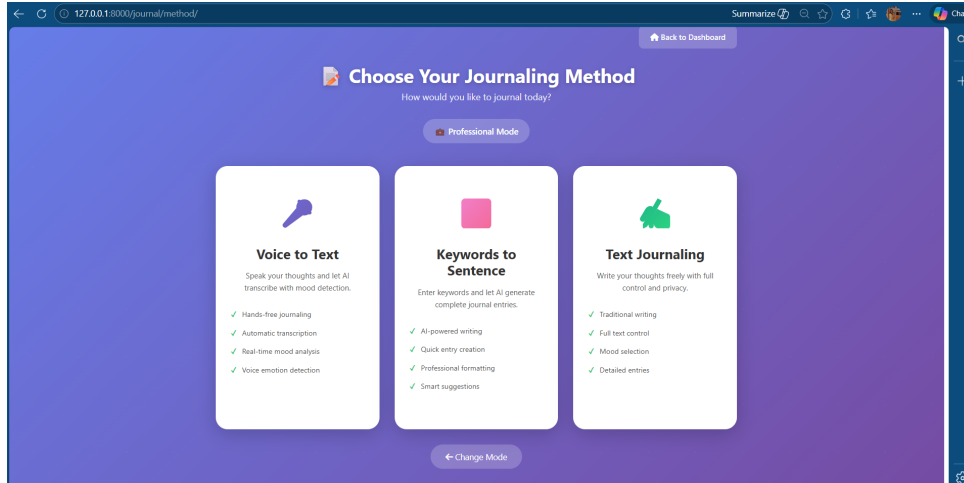


Figure 4.5: Interface showing available journaling methods

4.4 Keyword-Based Journaling

The keyword journaling module allow users to enter keywords which represent their thoughts or experience. The system then generate meaningful sentences based on given keywords and selected tone.

Figure 4.6: Keyword input interface for journaling

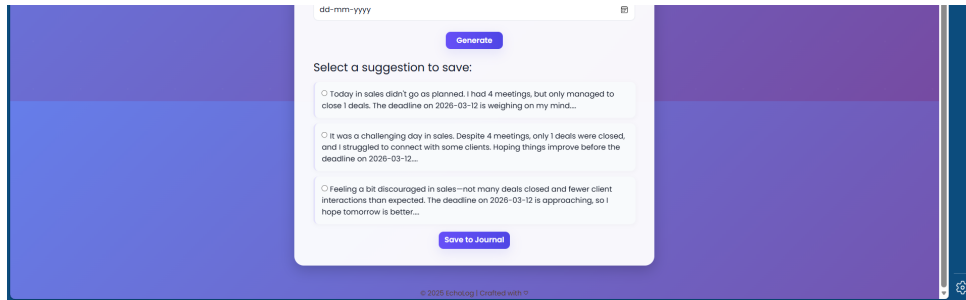


Figure 4.7: Keyword suggestion feature to help user during journaling

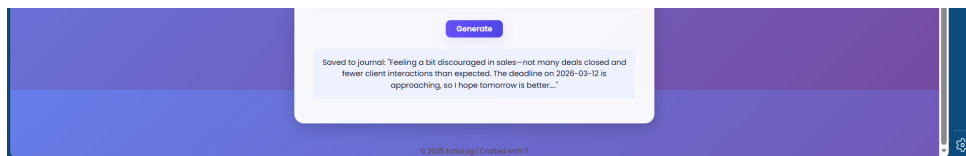


Figure 4.8: Generated journal entry saved after keyword processing

4.5 Text-Based Journaling

Users can also enter journal entries manually using text journaling interface. This method allow users to express their thoughts freely without restriction.

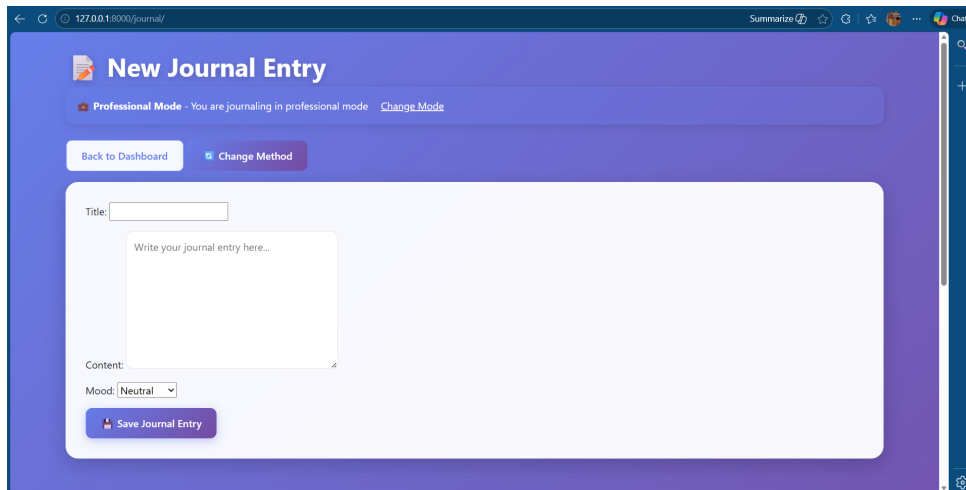


Figure 4.9: Text journaling interface for entering journal content

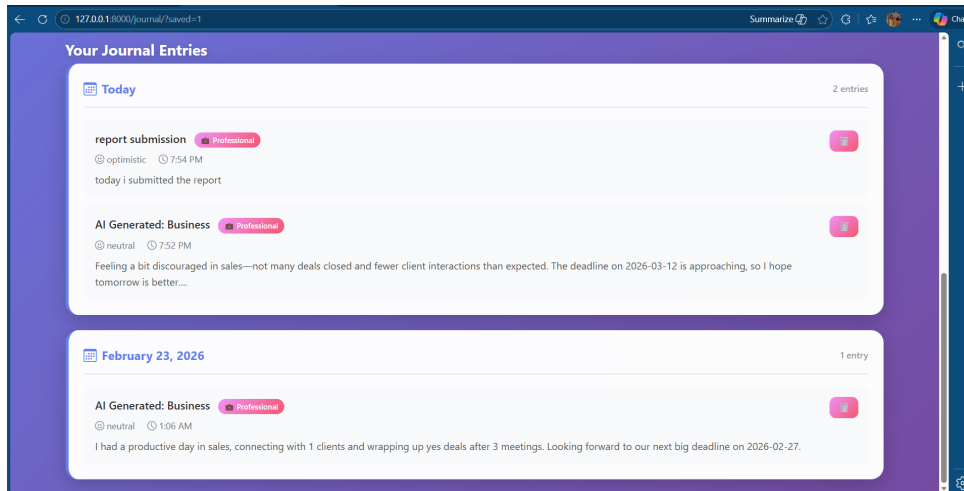


Figure 4.10: Recently stored journal entries display

4.6 Voice-to-Text Journaling

Voice journaling feature allow users to record thoughts using speech input. The system convert spoken words into text using Whisper speech recognition model.

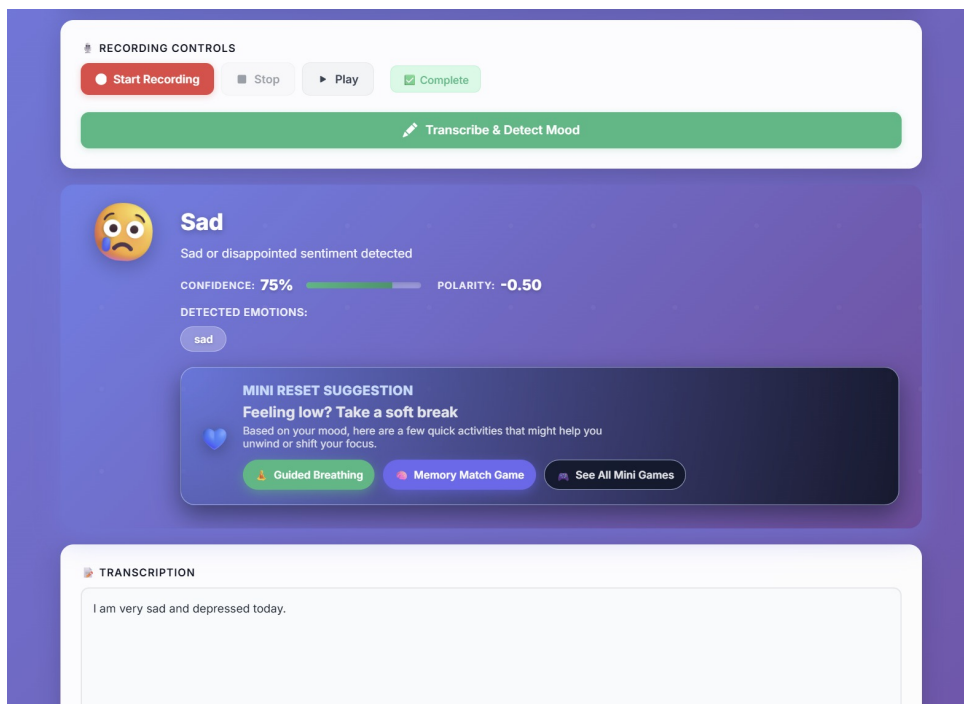


Figure 4.11: Voice to text journaling interface using speech recognition

4.7 Task Management Features

EchoLog also include productivity tools such as to do list manager. Users can add tasks and track daily activities directly from dashboard.

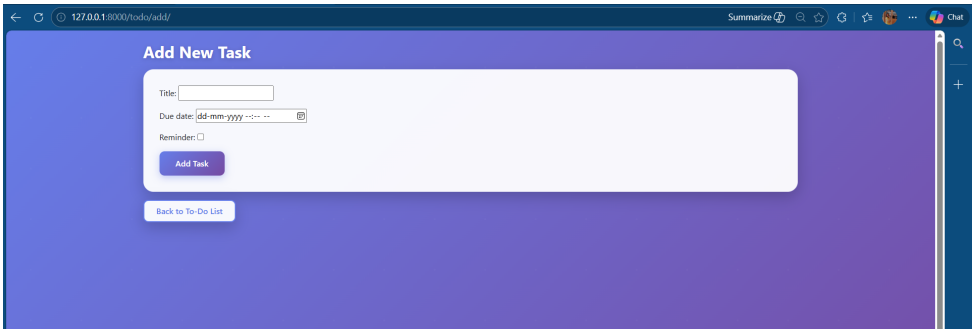


Figure 4.12: Interface for adding tasks to to-do list

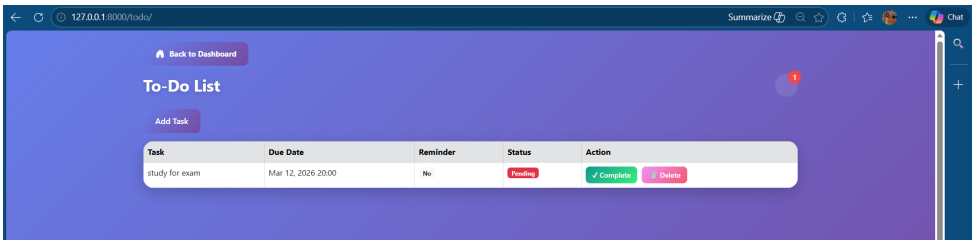


Figure 4.13: Task added and displayed in task list

4.8 Reminder and Notification System

The reminder system allow users to schedule journaling reminder and task notification. These reminders help users to maintain regular journaling habit.

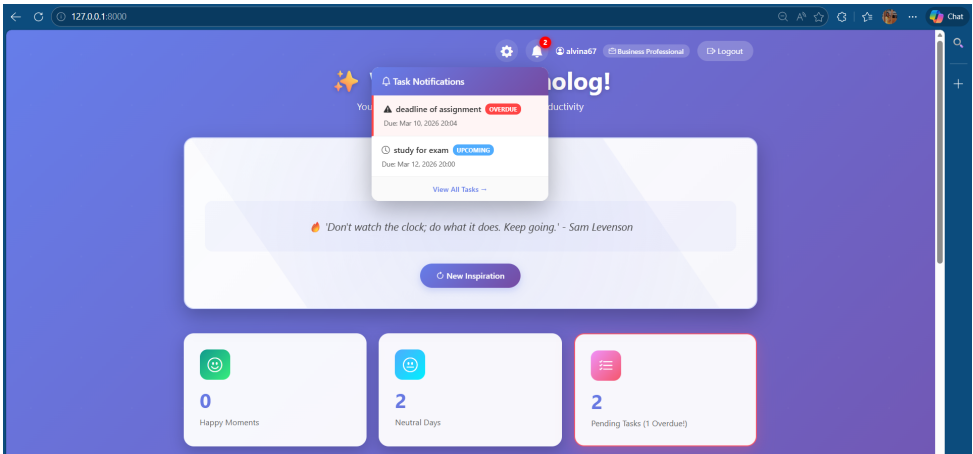


Figure 4.14: Reminder configuration interface

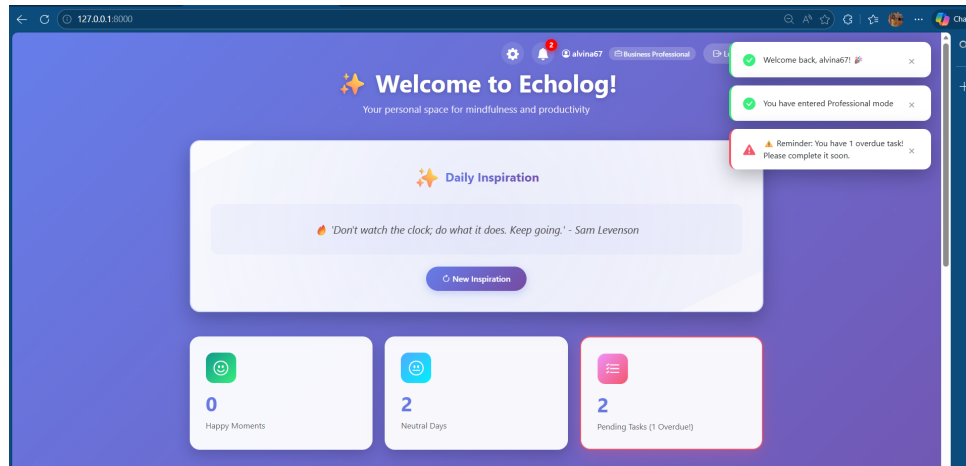


Figure 4.15: Reminder notification shown to user

4.9 Stress Relief and Mini-Games

To improve user engagement and reduce stress, EchoLog system include simple mini games which are suggested when negative emotion is detected.

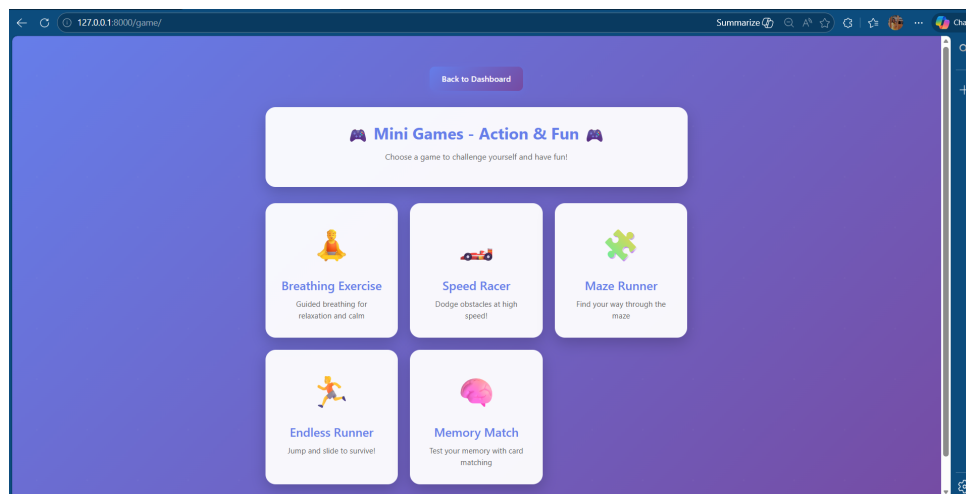


Figure 4.16: Mini games interface for engagement

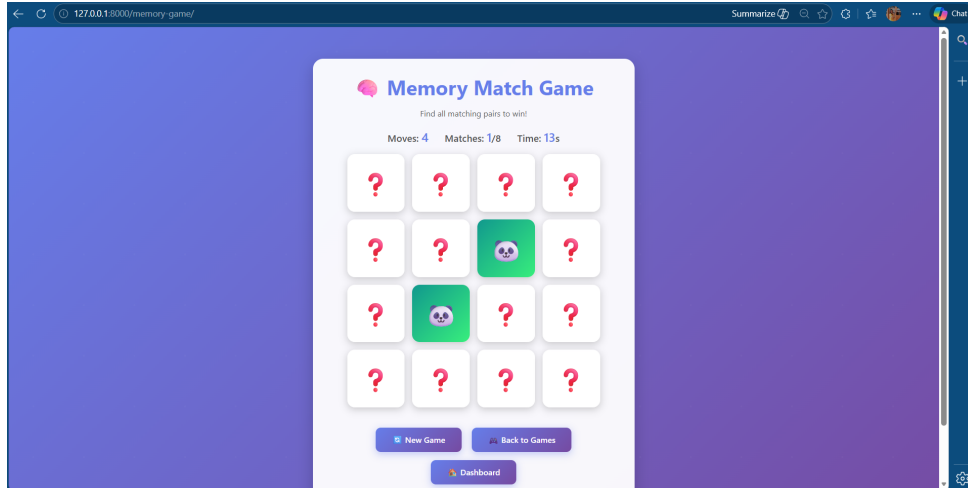


Figure 4.17: Memory game for focus and relaxation

4.10 System Performance Evaluation

4.10.1 Voice-to-Text Accuracy

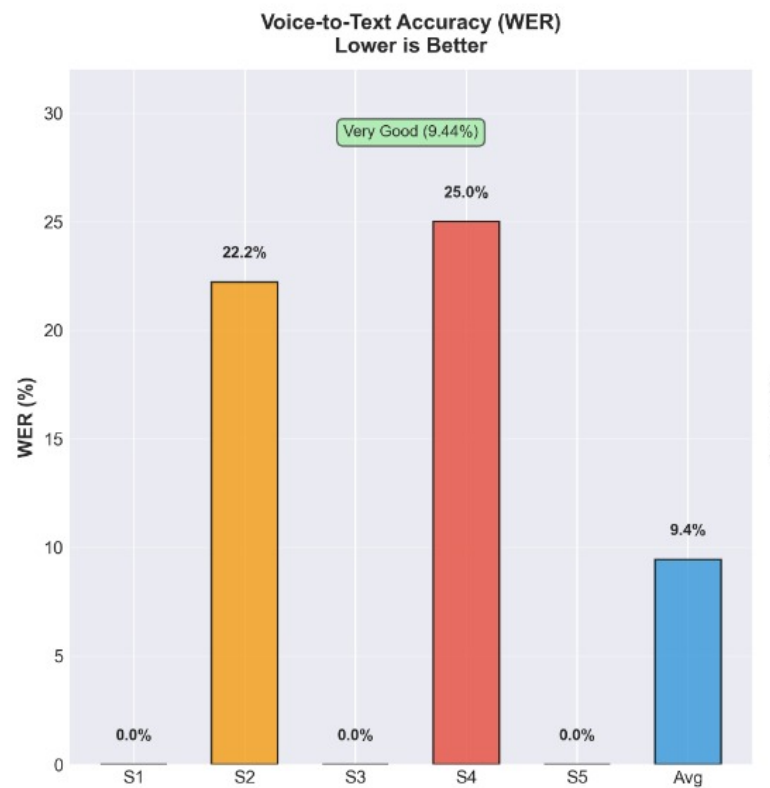


Figure 4.18: Voice to text accuracy evaluation

The figure 4.18 shows the performance of the voice-to-text module used in the EchoLog system. The Word Error Rate (WER) values indicate how accurately the system converts spoken input into text format. Lower WER values represent better transcription accuracy and more reliable results. The experimental evaluation shows that most test samples achieved very low error rates. The overall average WER of approximately 9.4% demonstrates that the Whisper-based speech recognition model performs effectively for voice journaling tasks.

4.10.2 Sentiment Analysis Performance

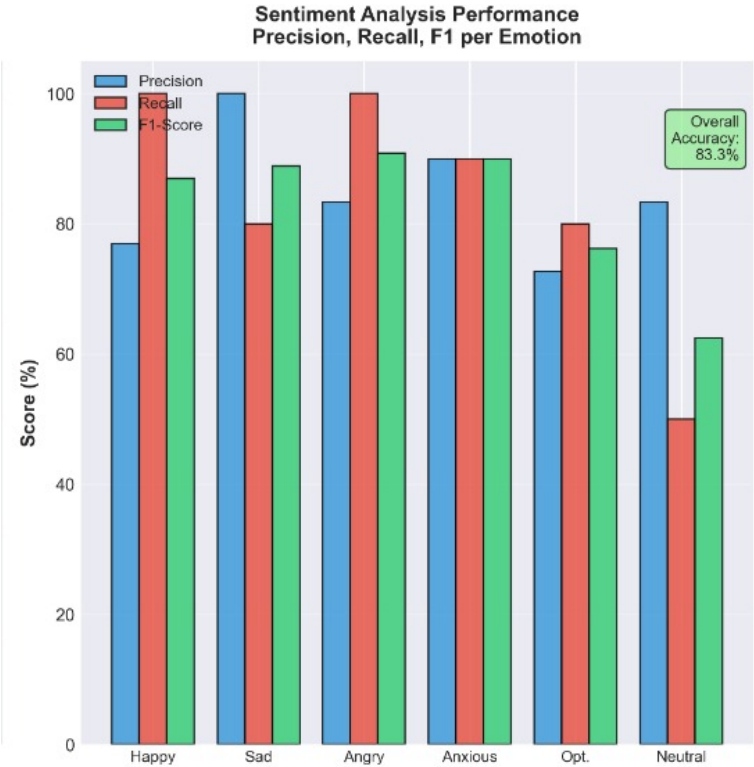


Figure 4.19: Precision, recall and F1 score for sentiment classification

The figure 4.19 represents the sentiment analysis performance of the EchoLog system across multiple emotional categories. The evaluation metrics include precision, recall and F1-score, which measure the accuracy of emotion classification. The model analyses journal entries and identifies emotions such as happy, sad, angry, anxious, optimistic and neutral. The results show consistently high scores across most emotional categories. The

overall accuracy of approximately 83.3% indicates that the sentiment analysis module can effectively detect emotional patterns in user journals.

4.10.3 User Experience Metrics

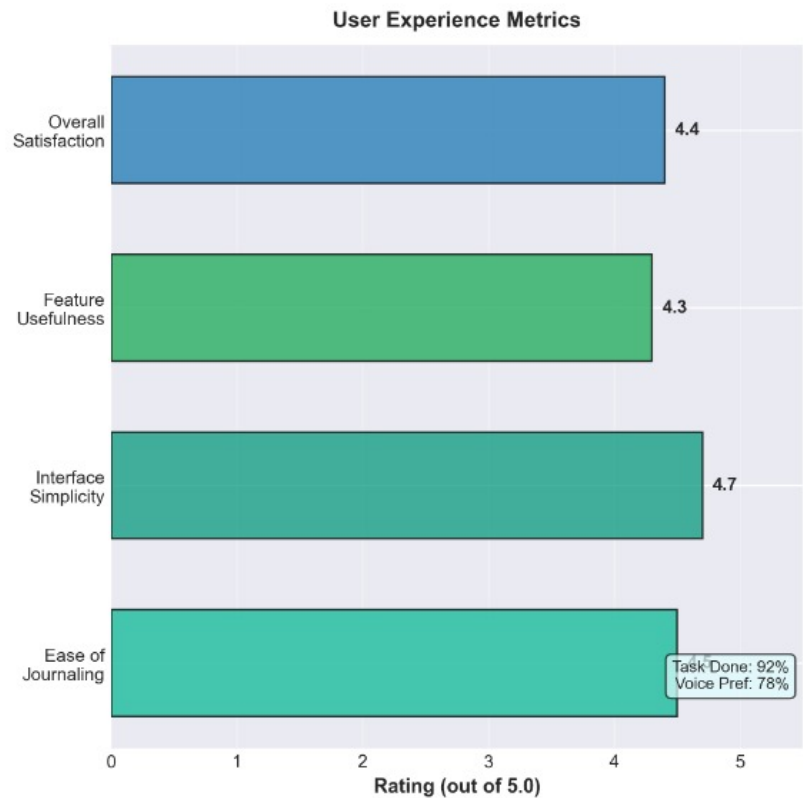


Figure 4.20: User experience evaluation metrics

The figure 4.20 represents the feedback collected from users after interacting with the EchoLog application. Several user experience factors were evaluated including ease of journaling, interface simplicity, feature usefulness and overall satisfaction. The ratings are measured on a scale of 1 to 5, where higher values indicate better user experience. Most features received ratings above 4, which shows that users found the system intuitive and helpful. These results indicate that the application interface and journaling features contribute positively to user engagement and usability.

4.10.4 System Performance Metrics

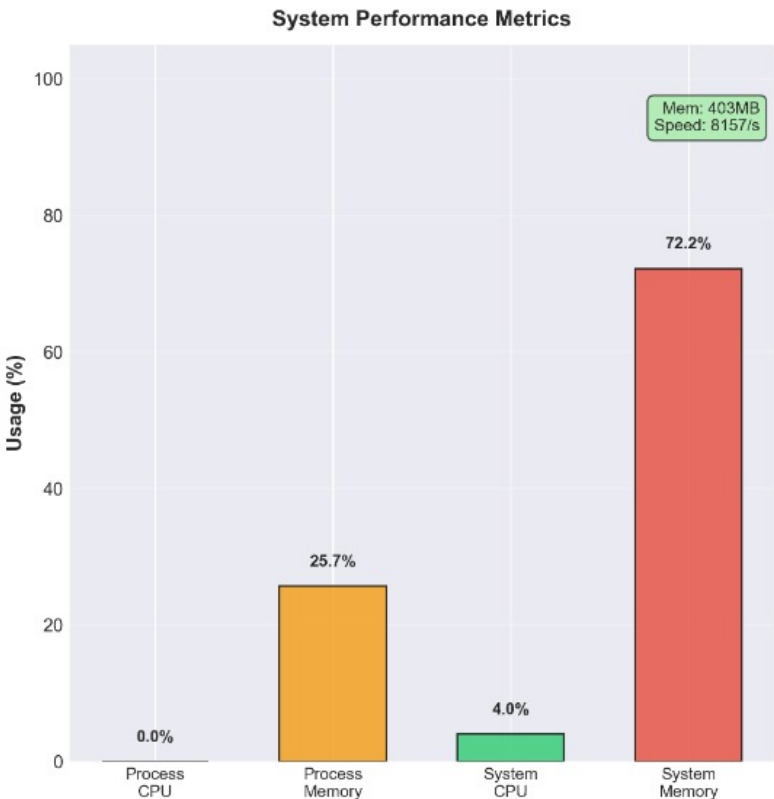


Figure 4.21: System performance metrics showing response time and efficiency

The figure 4.21 illustrates the system performance in terms of CPU usage, memory consumption and processing efficiency. The results indicate that the application requires only minimal CPU resources during operation. The memory usage remains moderate even while performing multiple tasks such as journaling, sentiment analysis and voice transcription. This demonstrates that the system is optimized for efficient resource utilization. As a result, the EchoLog application can run smoothly without placing heavy load on the device.

Chapter 5

Conclusions & Future Scope

This chapter summarize the outcomes of the project “EchoLog: An AI-Powered Digital Journaling System”. The main goal of this project was to design and develop an intelligent journaling platform which enables users to record their daily thoughts and experiences using both text and voice input while also giving emotional insights. The system integrate Artificial Intelligence and Natural Language Processing techniques to analyse journal entries and identify emotions and topics expressed by the user. Through this project, the team successfully showed how AI can improve traditional journaling by giving automatic analysis and reflections.

The project mainly focused on the design, development and implementation of the core components of the journaling platform. Technologies such as Whisper was used for speech to text conversion, which allow users to create journal entries through voice input. Sentiment analysis and emotion detection was performed using NLP techniques including TextBlob and transformer based models like DistilBERT. A simple and user friendly interface was also developed so that users can easily create, store and analyse their journal entries. The system show how AI based tools can be integrated to support self reflection, emotional awareness and personal growth.

In future, the system can be improved by integrating more advanced emotion detection models to increase the accuracy and better understanding of journal entries. Features like mood trend visualization, personalised insights and intelligent recommendation can be added to give deeper analysis of user emotions over time. The platform can also be expanded to support different types of users such as students, working professionals and elderly peoples who may get benefit from maintaining digital journals for emotional expression and memory tracking.

Also the system can be developed as a mobile application with secure cloud storage to improve accessibility and usability, and also to maintain better data security.

References

1. H. Li, K. Shi, D. Wang, X. Wang, and G. Xu, “A topic-controllable keywords-to-text generator with knowledge base network,” *CAAI Transactions on Intelligence Technology*, 2024.
2. Y. Sasazawa, T. Morishita, H. Ozaki, O. Imaichi, and Y. Sogawa, “Controlling keywords and their positions in text generation,” in *Proc. International Natural Language Generation Conference (INLG)*, 2023.
3. Y. Liu, X. Yang, and D. Qu, “Exploration of Whisper fine-tuning strategies for low-resource ASR,” *EURASIP Journal on Audio, Speech, and Music Processing*, 2024.
4. S. Rijal, S. Adhikari, M. Dahal, M. Awale, and V. Ojha, “Whisper finetuning on Nepali language,” *arXiv preprint*, 2024.
5. A. K. Sharma, M. R. Pandya, and A. Shukla, “Fine-tuning Whisper Tiny for Swahili automatic speech recognition,” in *Proc. AfricaNLP Conference*, 2025.
6. D. Trihapningsari, A. Widyasuri, M. A. Putri, and A. Fatihin, “Sentiment analysis of ChatGPT exploration based on opinions on platform X using Naïve Bayes algorithm,” in *Proc. International Seminar on Science and Technology (ISST)*, 2025.
7. R. F. P. Pratama and W. Maharani, “Comparative analysis of Naïve Bayes and SVM for improved emotion classification on social media,” *Edumatic Journal*, 2025.
8. B. Zhang, H. Yang, and X.-Y. Liu, “Instruct-FinGPT: Financial sentiment analysis by instruction tuning of large language models,” *arXiv preprint*, 2023.
9. N. S. Azwarni *et al.*, “Evaluating TextBlob, lexicon, SVM, Naïve Bayes and other techniques for sentiment analysis in stock market context,” *Journal of Theoretical and Applied Information Technology*, vol. 102, no. 13, 2024.

10. A. Vashisth, M. Kumari, and A. Mishra, “MoodSync: An AI-Powered Journal for Enhanced Emotional Well-being,” in *Proc. IEEE Global AI Summit*, 2024. [Online]. Available: <https://doi.org/10.1109/GLOBALAISUMMIT62156.2024.10947968>

Appendix A: Presentation

EchoLog: An AI Journal

100 % Project Evaluation

Presented by
Group 8

Guided by:

Shini Sam

Assistant Professor

DCS

RSET

Group Members:

Alvina Sheen T (U2203033)

Basil Raju (U2203064)

Adarsh S (U2203016)

CONTENTS

- Problem Definition
- Project Objective
- Purpose and Need
- Literature Survey
- Proposed Method
- Architecture Diagram
- Other design diagrams
- Module description
- Module wise diagrams
- Results and Outputs
- Assumptions
- Work breakdown and responsibilities
- Hardware and Software Requirements
- Gantt Chart
- Risks and Challenges
- Conclusion
- References

PROBLEM DEFINITION

In today's fast-paced digital world, people struggle to maintain consistent journaling habits due to time constraints and writing effort. Traditional journaling apps only store text and do not understand emotions or context. They lack intelligent sentence generation, voice integration, and emotional support features. There is no unified system that combines AI-based mood detection, keyword sentence generation, and interactive wellness support.

PROBLEM OBJECTIVE

The objective of EchoLog is to develop an AI-powered journaling system that converts voice input into text, generates meaningful sentences from user-provided keywords, and supports tone-based sentence generation. The system detects user mood using NLP and transformer-based models, offers both personal and professional journaling modes, triggers stress-relief games when sadness is detected, and integrates to-do lists with reminders to enhance productivity and emotional well-being.

PURPOSE AND NEED

The purpose of EchoLog is to transform traditional journaling into an intelligent and emotionally supportive experience. Many users find it difficult to consistently maintain journals due to time constraints and writing effort. There is a need for a system that reduces manual effort, understands user emotions, and provides personalized assistance. EchoLog fulfills this need by integrating AI, voice input, and mood detection into a single platform.

LITERATURE SURVEY

Paper 1 – Li He; Kaize Shi et al. (2024) [1]

Topic-Controllable Keywords-to-Text (TC-K2T) Model

- What the Paper Says
 - The paper proposes a model called TC-K2T.
 - It converts unordered keywords into meaningful sentences.
 - It keeps the generated text related to a specific topic.
 - It uses Knowledge Base Network (KBN) for topic understanding and Knowledge-guided Conditional Variational Autoencoder (KCVAE) for natural sentence generation
- Advantages
 - Maintains strong topic consistency
 - Produces natural and meaningful sentences
 - Useful for AI journaling and storytelling systems
 - Better than normal keyword-to-text generators
- Disadvantages
 - Model is complex (multiple components like KBN and KCVAE)
 - Requires large training data and knowledge base
 - May need high computational power
 - Hard to implement for small-scale projects

Paper 2 – Y. Liu; X. Yang; D. Qu (2024) [2]

Exploration of Whisper Fine-Tuning Strategies for Low-Resource ASR

- What the Paper Says
 - Studies fine-tuning methods for Whisper.
 - Focuses on improving Automatic Speech Recognition (ASR) for low-resource languages.
 - Compares different training strategies.
 - Aims to reduce Word Error Rate (WER).
 - Shows that proper fine-tuning improves speech-to-text accuracy.
- Advantages
 - Improves ASR accuracy significantly.
 - Reduces Word Error Rate.
 - Useful for voice-based AI systems.
 - Practical for real-world applications.
- Disadvantages
 - Requires labeled speech data.
 - Needs high computational resources (GPU).
 - Performance depends on language and accent.
 - Sensitive to background noise.

Paper 3 – N. S. Azwarni et al. (2024) [3]

Evaluating TextBlob, Lexicon, SVM, Naive Bayes and Other Techniques for Sentiment Analysis

- What the Paper Says
 - Compares different sentiment analysis techniques.
 - Evaluates TextBlob, Lexicon-based methods, Support Vector Machine (SVM), and Naive Bayes.
 - Focuses on stock market text data.
 - Measures accuracy and classification performance.
 - Identifies the most effective technique.
- Advantages
 - Provides clear comparison of multiple methods.
 - Helps choose the best sentiment analysis model.
 - Useful for text classification tasks.
 - Includes both simple and advanced techniques.
- Disadvantages
 - Limited to stock market domain.
 - Accuracy depends on dataset quality.
 - Lexicon methods may miss context.
 - Traditional ML models require feature engineering.

Paper 4 – A. Vashisth, M. Kumari, and A. Mishra (2024) [4]

MoodSync: An AI-Powered Journal for Enhanced Emotional Well-being

- What the Paper Says
 - It focuses on analyzing written journal entries
 - It uses NLP with integrated ChatGPT LLM
 - Provides personalized suggestions based on detected emotions
 - The goal is improving emotional well-being through intelligent journaling
- Advantages
 - Helps users understand their emotions clearly
 - Provides personalized emotional support
 - Encourages regular journaling habits and help in mental health monitoring
 - Useful for stress management and self-reflection
- Disadvantages
 - Accuracy depends on quality of emotion detection model
 - May not fully understand complex human emotions
 - Privacy concerns with storing personal journal data
 - Requires continuous AI model updates

PROPOSED METHOD

- Accept user input through text or voice
- Convert voice input into text and detects the mood
- Generate tone-aware sentences from keywords
- Analyze text using DistilBERT and TextBlob for mood detection
- Classify emotion into categories (happy, sad, stressed, neutral)
- Trigger stress-relief game if negative mood is detected
- Display and store final journal entry





MODULE DESCRIPTION

Module 1: NLP & Keyword based Generation Module

Focus: Keyword-based sentence generation, prompt formulation, tone selection.

Responsibilities:

- Design and implement the Keyword-to-sentence generation engine.
- Design and implement Tone-based sentence generator (Formal, Cheerful, Sad, etc.).

Tools/Libraries:

- Django REST APIs
- Templates for different tone outputs (engine.py)

MODULE DESCRIPTION

Module 2: AI/ML & Module

Focus: Voice to text and mood detection.

Responsibilities:

- Build Voice-to-text transcription using Whisper.
- Build mood detection using DistilBERT and TextBlob.
- Handle audio input API (getUserMedia , MediaRecorder) and convert to text.

Tools/Libraries:

- Whisper (OpenAI)
- TextBlob and DistilBERT for mood detection

MODULE DESCRIPTION

Module 3: UI + System Integration, games, database

Focus: Frontend development, feature integration, games, and CRUD

Responsibilities:

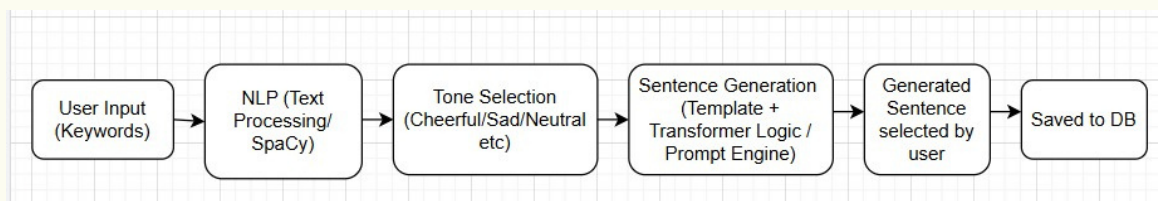
- UI integration and create full web UI using HTML, CSS, JavaScript.
- Develop pages: login, journal entry, to-do list, dashboard, games, etc.
- Build Django routes for handling backend logic.
- Handle to-do list with reminders and task management.

Tools/Libraries:

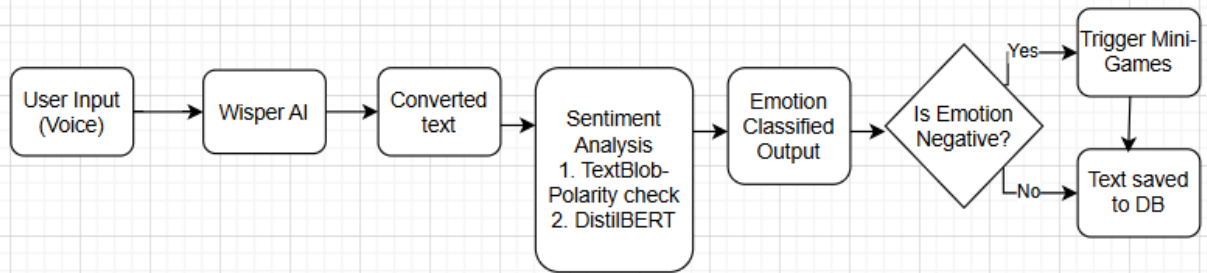
- Python Django for backend
- HTML5, CSS3, JavaScript for frontend
- SQLite for database

MODULE WISE DIAGRAMS

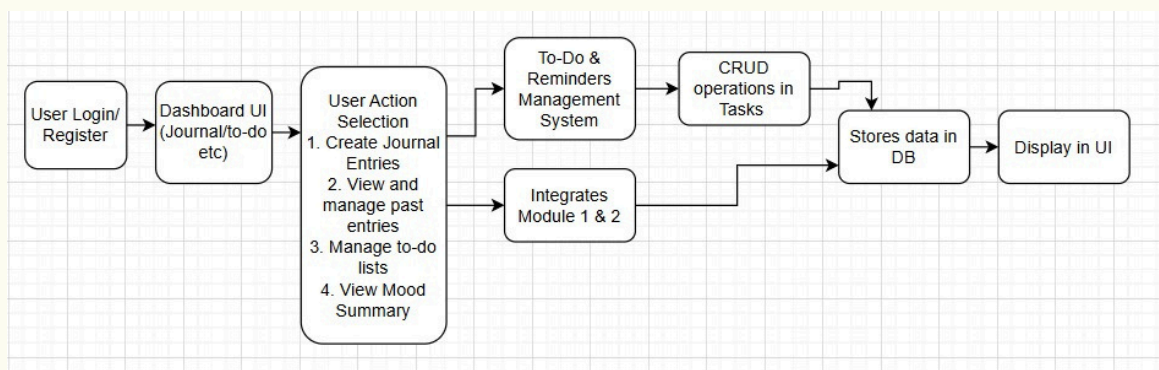
MODULE 1



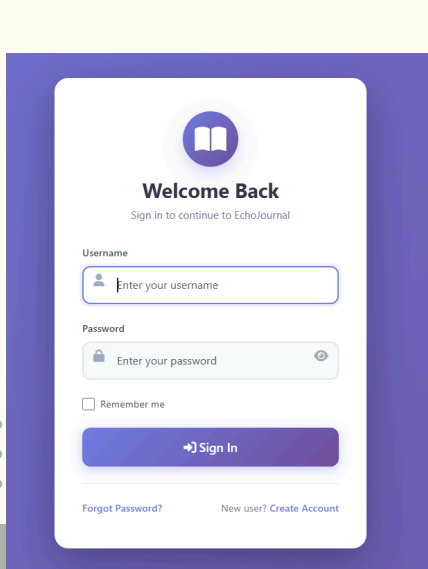
MODULE 2



MODULE 3



RESULTS AND OUTPUTS



Welcome Back
Sign in to continue to EchoJournal

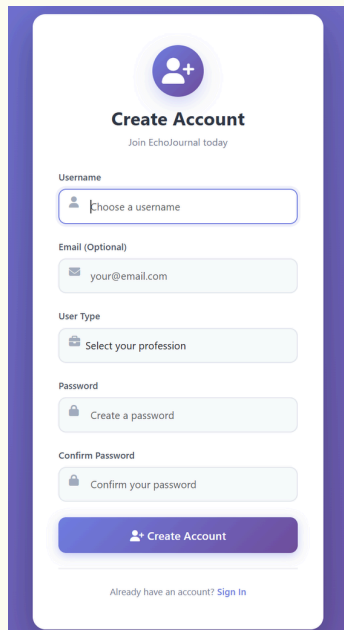
Username

Password

☐ Remember me

[Sign In](#)

[Forgot Password?](#) [New user? Create Account](#)



Create Account
Join EchoJournal today

Username

Email (Optional)

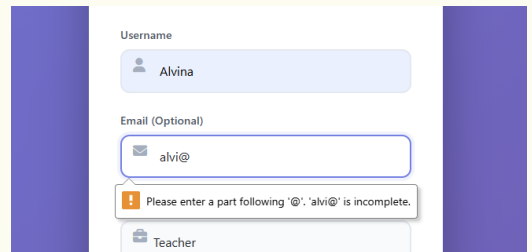
User Type

Password

Confirm Password

[Create Account](#)

[Already have an account? Sign In](#)

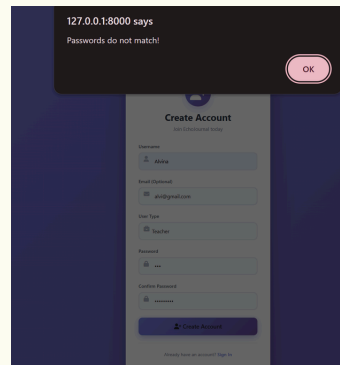


Username

Email (Optional)

Please enter a part following '@'. 'alvi@' is incomplete.

Teacher



127.0.0.1:8000 says
Passwords do not match!

[OK](#)

Create Account
Join EchoJournal today

Username

Email (Optional)

User Type

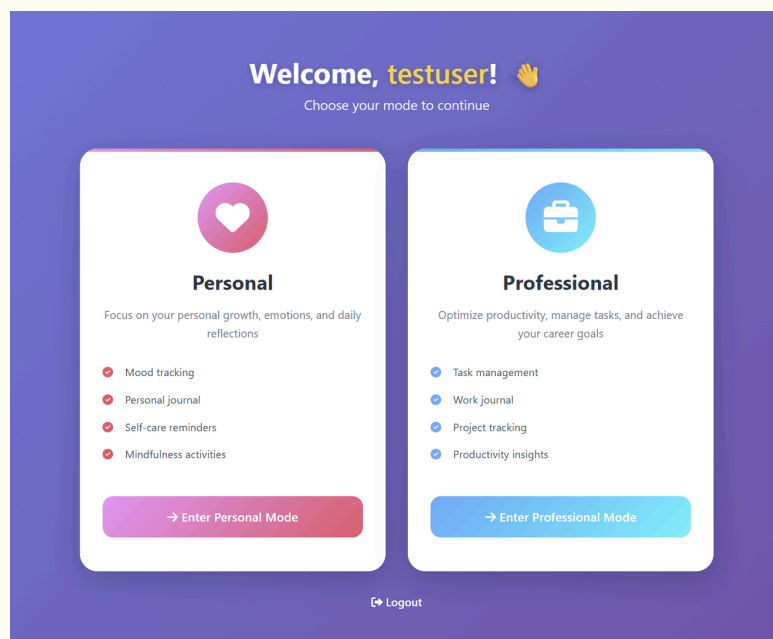
Password

Confirm Password

[Create Account](#)

[Already have an account? Sign In](#)

RESULTS AND OUTPUTS



Welcome, testuser! 🙌
Choose your mode to continue

Personal
Focus on your personal growth, emotions, and daily reflections

- Mood tracking
- Personal journal
- Self-care reminders
- Mindfulness activities

[Enter Personal Mode](#)

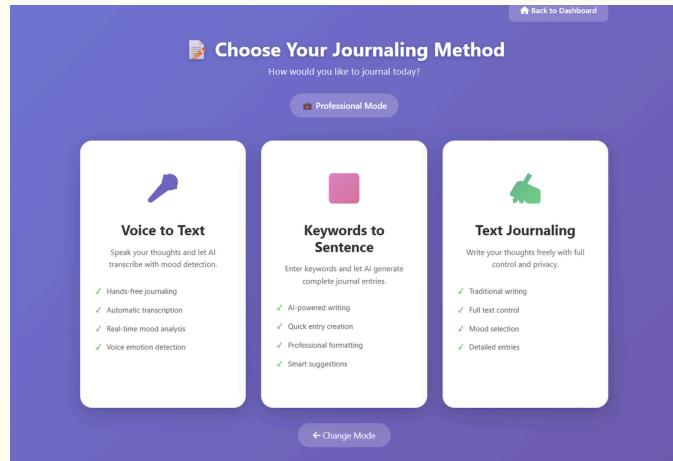
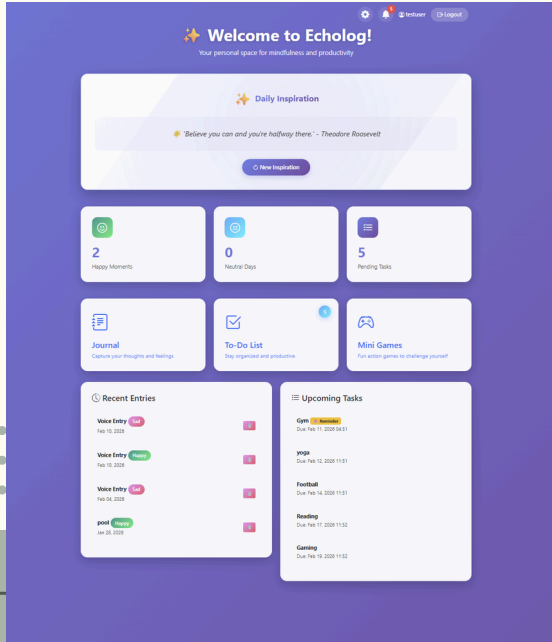
Professional
Optimize productivity, manage tasks, and achieve your career goals

- Task management
- Work journal
- Project tracking
- Productivity insights

[Enter Professional Mode](#)

[Logout](#)

RESULTS AND OUTPUTS



RESULTS AND OUTPUTS

Professional Mode - Your journal entries will be saved in professional mode
[Change Mode](#)

AI Journal Entry Generator

[Back to Dashboard](#)

Profession: Tone:

Subject taught:

Present Students:

Absent Students:

Assignments Given:

Homework Due Date:

[Generate](#)

© 2025 Echolog | Crafted with

Select a suggestion to save:

- ☐ What a fun day teaching maths! It was wonderful to see 77 students, though we missed 3. Homework is ready for 2026-02-13.
- ☐ Smiles all around in maths today! 77 attended, 3 missed out, and homework is set for 2026-02-13.
- ☐ I really enjoyed our maths class today—77 students joined in, and we hope 3 can be with us next time. Assignments are due on 2026-02-13.
- ☐ My maths students brought so much energy! 77 were present, 3 were missed. Looking forward to their homework by 2026-02-13.
- ☐ Feeling thankful for a lively maths class: 77 present, 3 absent. Excited to check assignments on 2026-02-13!

[Save to Journal](#)

Saved to journal: "What a fun day teaching ! It was wonderful to see students, though we missed . Homework is ready for ."

RESULTS AND OUTPUTS



Professional Mode - Your journal entries will be saved in professional mode
[Change Mode](#)

AI Journal Entry Generator

[Back to Dashboard](#)

Profession: Tone:

Subject taught:

Present Students:

Absent Students:

Assignments Given:

Homework Due Date:

[Generate](#)

[Generate](#)

Select a suggestion to save:

- ☐ Missing the usual energy in maths class. 3 students were absent, and assignments are due 2026-02-13...
- ☐ Today felt a bit quiet in maths—only 55 showed up, and we missed 3. Assignments are still due on 2026-02-13...
- ☐ It was a tough day with many absentees in maths. 55 attended, 3 were missing. Homework deadline is 2026-02-13...
- ☐ Sad to see so many absent in maths. Assignments are due on 2026-02-13...
- ☐ Low spirits in maths today with 3 missing. Hoping for better attendance next time. Homework due 2026-02-13...

[Save to Journal](#)

Saved to journal: "It was a tough day with many absentees in maths. 55 attended, 3 were missing. Homework deadline is 2026-02-13..."



RESULTS AND OUTPUTS



Voice to Text with Mood Detection
Record, transcribe, and analyze your emotional tone in real-time

[Dashboard](#)

[Save to Journal](#) [Clear All](#)

RECORDING CONTROLS

[Start Recording](#) [Stop](#) [Play](#) [Idle](#)

OR UPLOAD AUDIO FILE

[Choose File](#) No file chosen

Supported formats: MP3, WAV, OGG, M4A • Maximum file size: 25MB

[Transcribe & Detect Mood](#)

TRANSCRIPTION

Your transcribed text will appear here...

Entry title (optional)

Professional Mode - Your voice entries will be saved in professional mode
[Change Mode](#)

Voice to Text with Mood Detection

[Dashboard](#)

[Save to Journal](#) [Clear All](#)

RECORDING CONTROLS

[Start Recording](#) [Stop](#) [Play](#) [Complete](#)

OR UPLOAD AUDIO FILE

[Choose File](#) No file chosen

Supported formats: MP3, WAV, OGG, M4A • Maximum file size: 25MB

[Transcribe & Detect Mood](#)

Happy

Very positive sentiment detected

CONFIDENCE: 100% [View Details](#) POLARITY: 0.99

TRANSCRIPTION

Hello, how are you?

Entry title (optional)



RESULTS AND OUTPUTS



 **New Journal Entry**

Personal Mode - You are journaling in personal mode [Change Mode](#)

[Back to Dashboard](#) [Change Method](#)

Title:

Write your journal entry here...

Content:

Mood: Neutral

[Save Journal Entry](#)

 **New Journal Entry**

Personal Mode - You are journaling in personal mode [Change Mode](#)

[Back to Dashboard](#) [Change Method](#)

Title:

today I cook a special meal for my family

Content:

Mood: Happy

[Save Journal Entry](#)

RESULTS AND OUTPUTS



 **Memory Match Game**

Find all matching pairs to win!

Moves: 0 Matches: 0/8 Time: 2s

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

[New Game](#) [Back to Games](#)

[Dashboard](#)

 **Memory Match Game**

Find all matching pairs to win!

Moves: 5 Matches: 1/8 Time: 17s

?	?	?	?
?	?	?	?
?	?		
?	?	?	?

[New Game](#) [Back to Games](#)

[Dashboard](#)

RESULTS AND OUTPUTS

[Back to Dashboard](#)

To-Do List

[Add Task](#)

Task	Due Date	Reminder	Status	Action
Gym	Feb 11, 2026 04:51	Remind me	Pending	✓ Complete 🗑 Delete
yoga	Feb 12, 2026 11:51	No	Pending	✓ Complete 🗑 Delete
Football	Feb 14, 2026 11:51	No	Pending	✓ Complete 🗑 Delete
Reading	Feb 17, 2026 11:52	No	Pending	✓ Complete 🗑 Delete
Gaming	Feb 19, 2026 11:52	No	Pending	✓ Complete 🗑 Delete

Add New Task

Title:

Due date:

Reminder: ☐

[Add Task](#)

[Back to To-Do List](#)

RESULTS AND OUTPUTS

[🏠](#) [⚙](#) [🔔](#) [👤](#) [🔒](#) [🚪](#) [🔒](#) [🚪](#)

Welcome to Echolog!

Your personal space for mindfulness and productivity

🌟 Daily Inspiration

🔥 "The future belongs to those who believe in the beauty of their dreams." - Eleanor Roosevelt

[🌟 New Inspiration](#)

2
Happy Moments

5
Neutral Days

5
Pending Tasks (1 Overdue)

Journal
Capture your thoughts and feelings

To-Do List
Stay organized and productive
🔴 1 overdue

Mini Games
Fun action games to challenge yourself

🕒 Recent Entries

📅 Today 5 entries

AI Generated: Teacher [🔗](#)

📅 Upcoming Tasks

Gym [🔔](#) [🗑](#)
Due: Feb 11, 2026 04:51

Welcome to Echolog!

Your personal space for mindfulness and productivity

🔔 Task Notifications

- 🔴 Gym [OVERDUE](#)
Due: Feb 11, 2026 04:51
- 🕒 yoga [UPCOMING](#)
Due: Feb 12, 2026 11:51
- 🕒 Football [UPCOMING](#)
Due: Feb 14, 2026 11:51
- 🕒 Reading [UPCOMING](#)
Due: Feb 17, 2026 11:52
- 🕒 Gaming [UPCOMING](#)
Due: Feb 19, 2026 11:52

[View All Tasks →](#)

ASSUMPTIONS

- Users provide meaningful keywords
- Internet connection is stable for AI processing
- Voice input is clear and understandable
- Emotion categories are limited to predefined classes
- Users interact ethically and responsibly

WORK BREAKDOWN & RESPONSIBILITIES

Member 1: Keyword-Based Journaling Module

- Developed keyword-to-sentence generation system
- Implemented tone-based sentence generation
- Built and integrated NLP backend logic

Member 2: Voice-to-Text & Mood Detection Module

- Implemented voice-to-text journaling using Whisper
- Developed emotion detection using DistilBERT and TextBlob
- Designed and integrated mood classification workflow

Member 3: System Integration, Games & Productivity Module

- Designed frontend UI and integrated backend modules
- Developed stress-relief mini-games
- Implemented to-do list, reminders, and database management

HARDWARE & SOFTWARE REQUIREMENTS

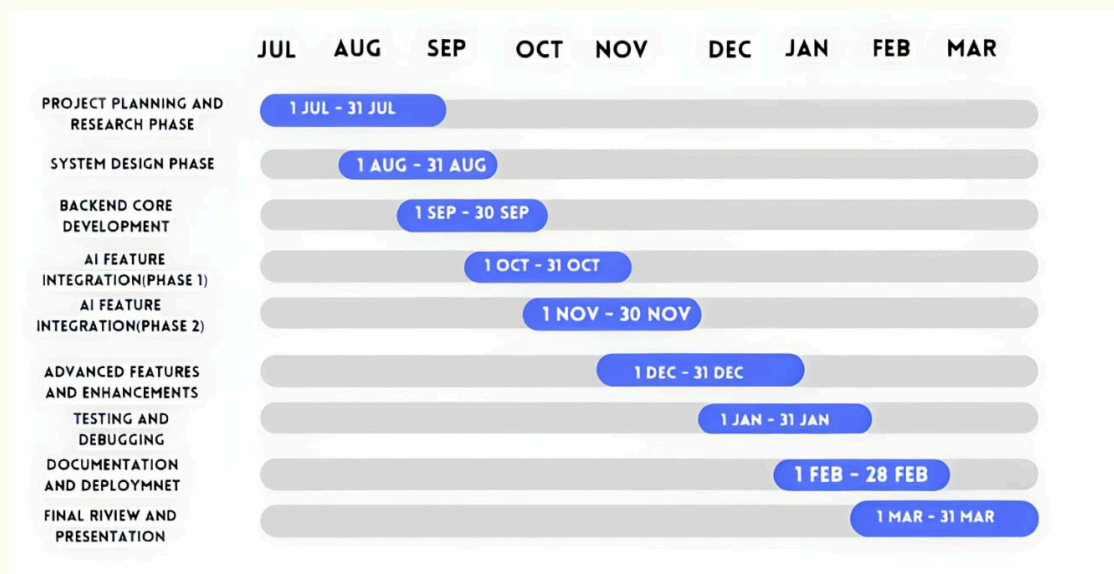
HARDWARE REQUIREMENTS

- Processor: Intel i5 or above
- RAM: Minimum 8GB
- Storage: 256GB or more
- Microphone of the device (for voice input)
- Laptop/PC

SOFTWARE REQUIREMENTS

- Python 3.10+
- Django Framework
- Whisper (OpenAI)
- TextBlob
- DistilBERT (HuggingFace Transformers)
- SpaCy
- HTML, CSS, JavaScript
- SQLite
- VS Code

GANTT CHART



RISKS AND CHALLENGES

- Inaccurate voice transcription due to noise
- Misclassification of emotions
- Privacy concerns for journal data
- AI-generated irrelevant sentences
- Limited computational resources

CONCLUSION

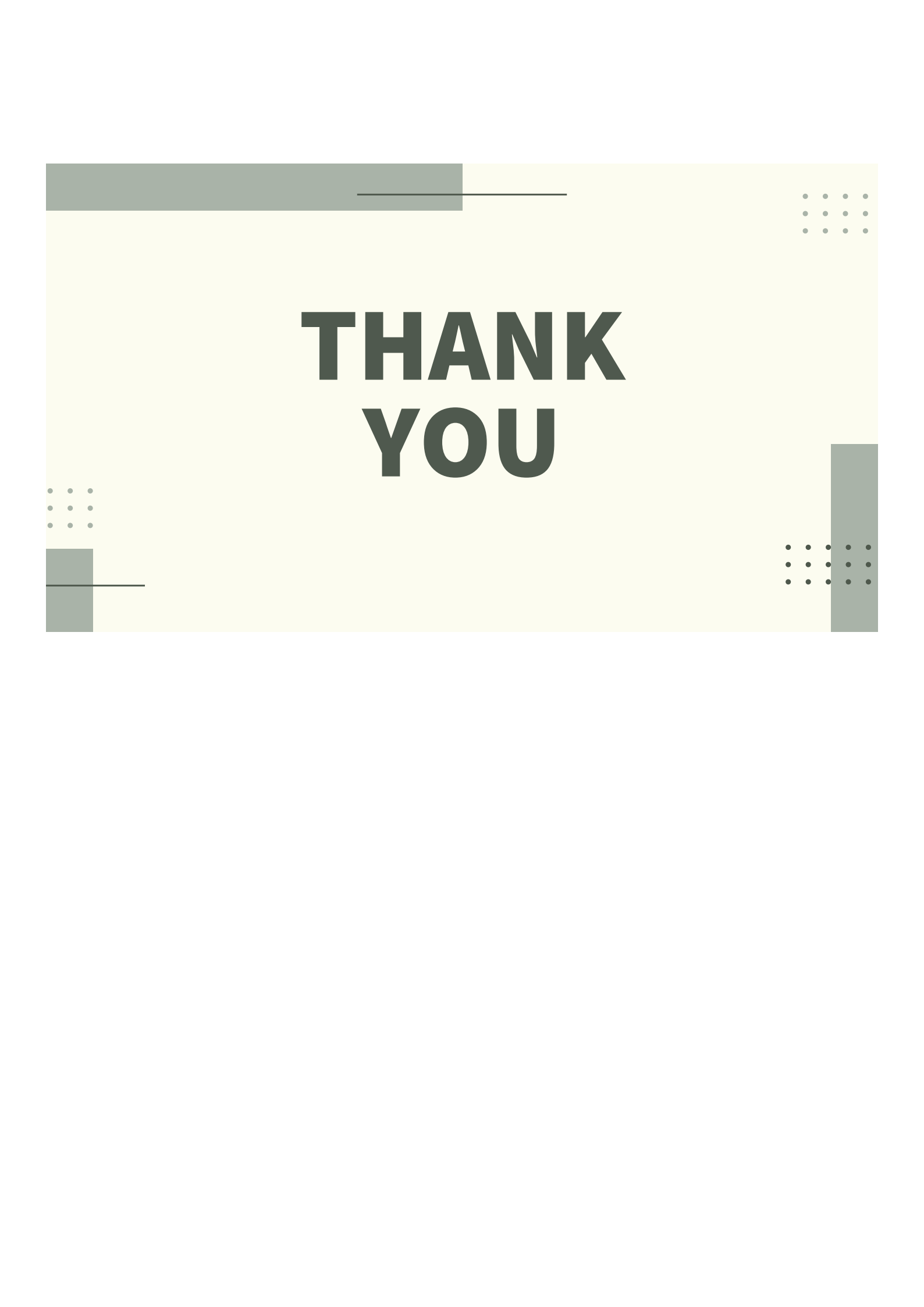
EchoLog is an AI-powered, emotion-aware journaling system that integrates voice-to-text transcription, keyword-based sentence generation, and mood detection into a unified platform. It reduces writing effort, improves emotional awareness, and promotes user engagement through intelligent interaction. By combining productivity features with emotional support mechanisms, EchoLog provides a modern and interactive journaling experience suitable for today's digital era.

REFERENCES

- [1] L. He, K. Shi, D. Wang, X. Wang, and G. Xu, "A topic-controllable keywords-to-text generator with knowledge base network," CAAI Trans. Intell. Technol., vol. 9, no. 3, pp. 585–594, 2024. [Online]. Available: <https://doi.org/10.1049/cit2.12280>
- [2] Y. Liu, X. Yang, and D. Qu, "Exploration of Whisper Fine-Tuning Strategies for Low-Resource ASR," EURASIP Journal on Audio, Speech, and Music Processing, vol. 2024, no. 29, 2024. doi: 10.1186/s13636-024-00349-3. [Online]. Available: <https://doi.org/10.1186/s13636-024-00349-3>
- [3] N. S. Azwarni, et al., "Evaluating TextBlob, Lexicon, Support Vector Machine, Naive Bayes and Other Techniques for Sentiment Analysis in Stock Market Context," Journal of Theoretical and Applied Information Technology, vol. 102, no. 13, pp. 4468–4479, Dec. 2024. [Online]. Available: <https://www.jatit.org/volumes/Vol102No13/20Vol102No13.pdf>
- [4] A. Vashisth, M. Kumari, and A. Mishra, "MoodSync: An AI-Powered Journal for Enhanced Emotional Well-being," in Proc. IEEE Global AI Summit, 2024. [Online]. Available: <https://doi.org/10.1109/GLOBALAISUMMIT62156.2024.10947968>
- [5] R.F.P. Pratama and W. Maharani, et al., "Comparative Analysis of Naive Bayes and SVM for Improved Emotion Classification on Social Media," Edumatic: Jurnal Pendidikan Informatika, vol. 9, no. 1, pp. 11–20, Apr. 2025. [Online]. Available: <https://ejournal.hamzanwadi.ac.id/index.php/edumatic/article/view/29087>

REFERENCES

- [6] B. Zhang, H. Yang, and X.-Y. Liu, "Instruct-FinGPT: Financial Sentiment Analysis by Instruction Tuning of Large Language Models," arXiv, 2023. [Online]. Available: <https://arxiv.org/abs/2306.12659>
- [7] A. K. Sharma, M. R. Pandya, and A. Shukla, "Fine-Tuning Whisper Tiny for Swahili ASR: Challenges and Recommendations for Low-Resource Speech Recognition," Proceedings of the Sixth Workshop on African Natural Language Processing (AfricaNLP 2025), Association for Computational Linguistics, pp. 74–81, Jul. 31, 2025. [Online]. Available: <https://aclanthology.org/2025.africanlp-1.11.pdf>
- [8] S. Rijal, S. Adhikari, M. Dahal, M. Awale, and V. Ojha, "Whisper Finetuning on Nepali Language," arXiv preprint arXiv:2411.12587, Nov. 19, 2024. [Online]. Available: <https://arxiv.org/abs/2411.12587>
- [9] D. Trihapningsari, A. Widyasuri, M. A. Putri, and A. Fatihin, "Sentiment Analysis of ChatGPT Exploration Based on Opinions on Platform X Using Naïve Bayes Algorithm," in Proc. 4th International Seminar of Science and Technology (ISST), vol. 4, pp. 009–015, Apr. 2025. [Online]. Available: <https://doi.org/10.33830/isst.v4i1.5235>
- [10] A. Vashisth, M. Kumari, and A. Mishra, "MoodSync: An AI-Powered Journal for Enhanced Emotional Well-being," in Proc. IEEE Global AI Summit, 2024. [Online]. Available: <https://doi.org/10.1109/GLOBALAISUMMIT62156.2024.10947968>

The card features a light yellow background. At the top left, there is a grey horizontal bar with a thin grey line extending to the right. At the top right, there is a 3x4 grid of small grey dots. In the center, the words 'THANK YOU' are written in a large, bold, dark grey sans-serif font. At the bottom left, there is a grey square with a thin grey line extending to the right. At the bottom right, there is a grey vertical bar with a 3x4 grid of small grey dots to its left.

**THANK
YOU**

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of

mathematics, natural sciences and engineering sciences

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and Team work: Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

10. Communication: Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation.

Make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

CO-PO AND CO-PSO MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 5.1: CO-PO-PSO Mapping

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.

Mapping	Level	Justification
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.

Mapping	Level	Justification
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.

Mapping	Level	Justification
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.

Mapping	Level	Justification
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.

Mapping	Level	Justification
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.